

Household Carbon Footprint & Target Tracker

Problem Statement Analysis

1. Problem Context & Overview

The assigned problem statement (PS #61, Sustainability & Green Tech track) calls for a green-living analytics tool. Households log their monthly utility consumption — electricity in kWh, gas usage, and commute distance in kilometres — and the system converts this into CO₂ equivalent (CO₂e) emissions. Beyond raw calculation, the tool also lets residents set carbon-reduction targets and track their progress toward those targets over time, turning a one-off calculator into an ongoing behavioural tool.

2. Stakeholders / Actors

- Resident — the primary user. Logs monthly consumption data, sets personal reduction targets, and monitors progress over time.
- Sustainability Coach — a secondary, advisory user. Reviews a Resident's logged data and historical trends, and provides written recommendations to help the Resident hit their targets.

Both actors are named explicitly in the problem brief; no additional actors were introduced, since inventing extra actors not implied by the scenario would misrepresent the assigned problem.

3. Requirement Elicitation Approach

Requirements were elicited by walking through the problem statement clause by clause and asking, for each noun phrase and verb, “what must the system do, and how well must it do it?” This produced two categories:

- Functional needs — concrete actions the system performs on behalf of an actor (ingesting logs, computing emissions, setting and tracking targets, notifying, and supporting coach review).
- Non-functional needs — quality constraints on those actions (response latency for charts, and data security/privacy given that utility and commute logs can reveal home-occupancy patterns).

4. Functional Requirements Identified

- FR-001 — Ingest monthly electricity and transport logs and compute total CO₂e emissions. (Given in the brief; the foundational capability everything else depends on.)
- FR-002 — Allow the Resident to set a monthly or annual carbon-reduction target. Without this, there is nothing to “track” — it operationalises the word “target” in the problem title.
- FR-003 — Calculate and display percentage progress toward the active target after each new log. This is the direct consequence of pairing logging (FR-001) with a target (FR-002).
- FR-004 — Notify the Resident when a progress milestone is reached. Inferred from the phrase “track reduction milestones” in the problem context — a milestone that is silently tracked but never surfaced would not meet user intent.
- FR-005 — Allow the Sustainability Coach to view household data and submit recommendations. Directly required by the Coach's presence as a named stakeholder — an actor with no use case would be a modelling error.

5. Non-Functional Requirements Identified

- NFR-001 — Given in the brief: the annual milestone chart must render interactive historical comparisons in under 200ms, under simulated peak load, meeting stated security standards.
- NFR-002 — Derived: household consumption data must be encrypted at rest and in transit, and scoped so only the owning Resident and their assigned Coach can access it. Added because utility and commute data are sensitive — they can reveal when a household is occupied — and the problem brief's mention of “security standards” in NFR-001 signalled that security was a first-class concern for this system, not just a performance afterthought.

6. Assumptions & Scope Boundaries

- Emissions conversion factors are assumed to follow EPA standards, per the acceptance criteria given in the sample FR-001.

- Only two actors are in scope, matching the brief; no billing, admin, or third-party integrations (e.g. smart meters) are assumed unless a future iteration of the problem statement adds them.
- The Sustainability Coach's recommendations are advisory text only — the problem statement does not require automated recommendation generation, so this was kept as a manual, human-in-the-loop action.

7. Traceability to the Use-Case Model

Each functional requirement maps to one or more use cases in the accompanying UML diagram. FR-001 splits into a base use case (Log Monthly Consumption) and an included use case (Compute CO2e Emissions), demonstrating the <<include>> relationship. FR-004's milestone notification is modelled as an optional <<extend>> of Track Progress Toward Target, since an alert only fires conditionally, not on every progress update — this is the textbook use of extend for optional/conditional behaviour. NFR-001's performance requirement motivated adding a dedicated View Historical Comparison Chart use case, even though no single FR names it directly.